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NMAM INSTITUTE OF TECHNOLOGY, NITTE
Off-Campus Centre of Nitte (Deemed to be University)
First Semester B.Tech (CBCS) Degree Examinations

December 2024

CS1005-2 – INTRODUCTION TO PYTHON PROGRAMMING
(For CV)

Duration: 3 Hours

Max. Marks: 100

Note:

Part – A: Multiple Choice Questions: Answer all **Twenty** questions in the **OMR Sheet** provided. Each question carries equal marks.

Part – B: Descriptive Answer Questions: Answer **Five full** questions choosing **Two full** questions from **Unit – I & Unit – II each** and **One full** question from **Unit – III**.

PART - A: MULTIPLE CHOICE QUESTIONS

20 Marks

- Which of the following is the first step in program development?
A) Writing the code
B) Problem analysis
C) Problem design
D) Debugging
- In a flowchart, parallelogram is used to represent _____
A) start and end point
B) process
C) input and output
D) decision
- Which of the following variable names is invalid in Python?
A) _name
B) name1
C) Name_
D) 1name
- What is the output of the following code?

```
for i in range (5) :  
    if i == 3 :  
        continue  
    print (i)
```


A) 0 1 2 3 4
B) 0 1 2 4
C) 1 2 4 5
D) 1 2 3 4
- How do you define a function in Python?
A) function my_func():
B) define my_func():
C) fun my_func()
D) def my_func():
- Which of the following statements is true about recursion in Python?
A) A function used for infinite loops
B) A function that can call itself
C) A function must always return None
D) Recursion is not allowed in Python
- Named arguments are also known as _____
A) Keyword arguments
B) Default arguments
C) Required arguments
D) Positional arguments
- What is the output for the following:

```
def func (a, b = 10, c = 20):  
    return a + b + c  
print (func (5, c = 15) )
```


A) 25
B) 30
C) 35
D) error
- What is a key characteristic of a set?
A) It is written within common brackets
B) It is always sorted
C) It does not allow duplicate elements
D) It has a key : value pair

10. What will be the output of this code?
`T = (1, 2, 3, 4)`
`print(T[2])`
 A) 1 B) 2
 C) 3 D) 4
11. What is the correct python code to display the last four characters of "Digital India"?
 A) `str[0:4]` B) `str[4:]`
 C) `str[-4:0]` D) `str[-4:]`
12. Which of the following is the correct syntax to create a dictionary in Python?
 A) `d = { 1 : "one", 2 : "two" }` B) `d = [1, "one" : 2, "two" ]`
 C) `d = [1 : "one", 2 : "two"]` D) `d = {1, "one" : 2, "two" }`
13. Which of the following statements is true about objects?
 A) Objects cannot have methods B) Objects are instances of classes
 C) Objects are defined using the def keyword D) Objects are static and cannot be changed
14. What mode is used to open a file for writing in Python?
 A) 'r' B) 'w'
 C) 'rw' D) 'x'
15. How can you write multiple lines of text to a file in Python?
 A) `file.writelines()` B) `file.write()`
 C) `file.writeall()` D) `file.addlines()`
16. How do you read a specific number of bytes from a file?
 A) `file.get(10)` B) `file.readline(10)`
 C) `file.read(10)` D) `file.readbytes(10)`
17. What type of error is raised when you use an undefined variable in Python?
 A) TypeError B) ValueError
 C) SyntaxError D) NameError
18. How many errors does the following code contain?
`x = 10`
`y = '20'`
`z = x + int(y)`
`print(z)`
 A) 0 B) 1
 C) 2 D) 3
19. What is the primary purpose of the Pandas library in Python?
 A) To perform numerical computations B) To handle data visualization
 C) To facilitate data manipulation and analysis D) To build web applications
20. What function is used to plot a pie chart in Matplotlib?
 A) `plt.plot_pie()` B) `plt.pie()`
 C) `plt.circle()` D) `plt.pie_chart()`

PART - B: DESCRIPTIVE ANSWER QUESTIONS

Unit – I		Marks	BT*	CO*	PO*
1.	a) Explain the classification of data types in python.	8	L*1	1	1
	b) Write a python program to print the factorial of a number.	8	L3	2	2
2.	a) Explain the following with an example for each. (i) list and tuple, (ii) set and dictionary.	8	L1	1	1
	b) Define function and explain the need of function in python.	8	L2	2	2
3.	a) Explain mutable and immutable variables with an example.	8	L2	1	2
	b) Explain recursive function and write a recursive program to print the countdown from five to zero.	8	L3	2	2

Unit – II

4. a) Explain string operators with examples and write a python program to print a string in reverse order. 8 L3 3 1
- b) Explain the three different methods to read contents from a file. Illustrate with a python program. 8 L2 4 4
5. a) Write a python program to input your residential address and count the number of digits, words, lowercase and uppercase letters in that address using suitable built-in functions. 8 L3 3 1
- b) Describe in detail the steps involved in handling data files in python. 8 L2 4 4
6. a) Write the output for the following codes:
- (i) `a = [10, 20, 30, 40, 50]`
`print(a.pop(3))`
- (ii) `fruits = ['mango', 'apple', 'orange', 'Grapes']`
`fruits.sort()`
`print(fruits)`
- (iii) `num = (10, 15, 20, 25, 30)`
`print(num.index(20))`
- (iv) `n = (2, 5, 7, 11, 13, 17, 23)`
`print(n[: : 2])` 8 L2 3 1
- b) Explain the concept of a class and an object in Object-Oriented Concepts. 8 L1 4 2

Unit – III

7. a) Explain the types of errors in python with suitable examples. 8 L2 5 3
- b) Write a python program to create a suitable graph for the runs scored by three Indian batsman from 2015 to 2022 as shown below:
- | Name \ Year | 2015 | 2016 | 2017 | 2018 | 2019 | 2020 | 2021 | 2022 |
|----------------|------|------|------|------|------|------|------|------|
| Virat Kohli | 623 | 739 | 1460 | 1202 | 1177 | 431 | 329 | 802 |
| Rohit Sharma | 815 | 564 | 1293 | 1030 | 1490 | 171 | 90 | 1249 |
| Shikhar Dhawan | 745 | 287 | 960 | 897 | 583 | 290 | 297 | 970 |
- Add the title, axes, legend, and marker for the graph. 8 L3 5 4
8. a) Explain exception handling in python using the function try-except-else-finally with an example. 8 L2 5 4
- b) Write a short note on Pandas in python programming. 8 L1 5 3

BT* Bloom's Taxonomy, L* Level; CO* Course Outcome; PO* Program Outcome
